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SOFTWARE **ENGINEERING**

M02 — Session 04: Requirements Baseline, Traceability & Review

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Checkpoint 1 happens now — after M02 / Session 04.

UPLOAD TODAY

Export the complete project ZIP and upload it to Moodle as Checkpoint 1.

Submission	Required coverage	Points
Checkpoint 1	M01–M02	20
Checkpoint 2	M01–M04	20
Checkpoint 3	M01–M06	20
Checkpoint 4	M01–M08	20
Checkpoint 5 — final project	M01–M10	20
Total	**10 modules**	**100**

MOODLE CHECKPOINTS**

Checkpoint 1
M01–M02

Checkpoint 2
M01–M04

Checkpoint 3
M01–M06

Checkpoint 4
M01–M08

Checkpoint 5
final project*

***final project submission - 1 week after our last class**

***Always export your project before leaving the lab. A Moodle checkpoint is not the same thing as a progress backup.**

Syllabus topic	Coverage
Project vision, lifecycle and feasibility	M01
Business, functional and non-functional requirements	M02
System functions and information flows	M03–M04
Reliability, availability, security and fault tolerance	M05
Physical infrastructure and deployment	M06
UML diagrams and design patterns	M03, M07–M08
CASE tools	Website modelling tools used in required diagram tasks
Project management and professional practice	M09
Testing methods and practical exercises	M10: test plan, six test cases and two manual walkthroughs

QuickBite project - criteria / requirements

1. **Work in teams of 3**

Submit one agreed project ZIP per team. Everyone should contribute and be able to explain the team's decisions.

2. **Each module is worth 10 points**

Each module has 5 criteria worth 0–2 points each, published with its instructions:

- 2 points: complete, clear and justified.
- 1 point: partly correct, but incomplete or unclear.
- 0 points: missing or fundamentally incorrect.

3. **Different answers can earn full marks**

We assess your reasoning, use of the project brief and consistency between your answers. There is no single ideal solution.

4. **Focus on required tasks**

Optional tasks are extra practice. Skipping them causes no penalty; completing them adds no points.

5. **Submit your work through Moodle**

Save your answers on the website, export the latest project ZIP and upload it by the assignment deadline. "Saved" on the website confirms storage - it is not a grade.

6. **Project total: 100 points**

M01 is worth 10 points across Sessions 1 and 2 together. The project contributes to the laboratory component;

Before Session 04 — what you already know

Session 03 gave you the requirements content. **Today we structure, validate and review it.**

M01.5 — Business objectives

Outcome + metric + target / direction.

Why the project should create value.

Session 03 — Requirements

Functional requirements → services / behaviour

Non-functional requirements → quality / operation

Constraints → limits on choices

Today — Baseline & review

Keep the same requirements.

Validate links created in S03.

Check user behaviour, objective coverage, clarity and testability.

Do not create a second requirements list — continue the same baseline.

Requirements register: functional example

Car rental example

ID and type	FR-02 / Functional
Requirement	The system shall reject a reservation that overlaps an existing confirmed reservation for the same vehicle.
Source	Car rental brief: double booking can happen.
Priority	Must
Acceptance criterion	With a confirmed rental from 10 to 12 May, a new request for the same vehicle from 11 to 13 May is rejected.

Acceptance criteria

An acceptance criterion states **observable evidence** that **a requirement has been met.**

Requirement: exclude vehicles under maintenance from search results.

Criterion: with one available vehicle and one vehicle marked under maintenance, a search for the same branch and period returns only the available vehicle.

Acceptance criteria - How do we know if requirements have been met?

Success, failure and boundary cases

A requirement is easier to validate when we can state what **observable result** would show that it is satisfied.

Success when we succeed :) Normal valid situation.	No overlapping confirmed reservation exists. → **an otherwise valid reservation is accepted.**
Failure …when we fail :) A rule or condition is violated.	An overlapping confirmed reservation exists. → **the new reservation is rejected.**
Boundary Tests behavior exactly at the edge of a rule.	One reservation ends exactly when another starts. → **accept if back-to-back reservations are allowed; otherwise reject.**

If you cannot describe observable evidence, the requirement may still be **too vague**.

NFR requirements - goals are not yet testable requirements

You already know NFRs. **Today ask whether the wording you wrote can actually be verified**.

Weak wording:

The search should be fast.

Clearer wording:

At least 95% of availability searches shall finish within 2 seconds with 100 concurrent users and 1,000 vehicle records.

Invented targets are teaching assumptions — label them explicitly.

Functional requirements - What must the system do?

FR-01	The system shall search available vehicles for a selected branch and rental period.
FR-02	The system shall reject a reservation that overlaps an existing confirmed reservation for the same vehicle.
FR-03	The system shall allow authorised branch staff to mark a vehicle as under maintenance.

Quality requirements: Car Rental examples

QPR-01 (Performance)	95% of searches finish within 2 seconds with 100 concurrent users and 1,000 vehicle records.
QA-01 (Availability)	The reservation service is available for at least 99.5% of each calendar month, including planned downtime.
QU-01 (Usability)	At least 4 of 5 first-time test participants complete the standard reservation flow within 3 minutes without help.

QuickBite - Required Task 6

Describe one important user interaction

DELIVERABLE

Describe one important user interaction.

WHERE TO SAVE

Write on this task card and click Save answer. It is copied into the module report and included in your project ZIP.

MINIMUM REQUIREMENTS

- Write one user-oriented scenario
- Include a normal outcome
- Include one failure OR boundary outcome
- Link it to your requirement IDs

COMPLETE WHEN Your saved work meets the checklist above. Check it yourself; the website does not grade correctness.

QuickBite - Required Task 7

Review objective coverage and requirement quality

DELIVERABLE

Review objective coverage and requirement quality.

WHERE TO SAVE

Write on this task card and click Save answer. It is copied into the module report and included in your project ZIP.

MINIMUM REQUIREMENTS

- Review the links created in Task 2
- Check BOTH M01.5 objectives are supported
- Identify gaps or unjustified links
- Review clarity + testability
- Revise unclear requirements and explain the change
- If no revision is needed, state what you checked and why

COMPLETE WHEN Both M01.5 objectives and requirement quality have been reviewed;

Acceptance criteria - User-oriented scenario - Task 6

Scenario: A customer wants to reserve an available vehicle for a selected branch and rental period.

Success when we succeed :) Normal **valid** situation.	Vehicle is available and the request satisfies the rules. → **reservation is accepted.**
Failure ...when we fail :) A rule or condition is **violated**.	overlap exists → **reject**
Boundary **Tests** behavior exactly **at the edge of a rule**.	periods touch exactly → **accept or reject reservation**

Task 6 requires ONE scenario with a normal outcome + ONE failure OR boundary outcome.

User-oriented scenario - one important interaction

A scenario puts selected requirements into user context. It complements the baseline; it does not replace requirement IDs.

Example 1 - Reservation overlap

USER + GOAL

Car Rental:
A customer attempts to reserve a vehicle for a selected rental period.

Link: FR-02
(The system shall reject a reservation that overlaps an existing confirmed reservation for the same vehicle.)

NORMAL OUTCOME

No confirmed reservation overlaps the requested period.

→ The reservation **is not rejected** because of an overlap.

FAILURE OUTCOME

A confirmed reservation already **overlaps** the requested period for the same vehicle.

→ The new reservation is **rejected**.

BOUNDARY CASE

One reservation ends exactly when another starts.

→ Apply the defined back-to-back reservation policy.

Task 6 requires ONE scenario with a normal outcome + ONE failure OR boundary outcome.

User-oriented scenario - one important interaction

A scenario puts selected requirements into user context. It complements the baseline; it does not replace requirement IDs.

Example 2 - Search for an available vehicle

USER + GOAL

Car Rental:
A customer wants to find vehicles available at a selected branch for a selected rental period.

Link: FR-01
(The system shall search available vehicles for a selected branch and rental period.)

NORMAL OUTCOME

Vehicles are available for the selected branch and period.

→ The system returns the available vehicles.

NON-NORMAL OUTCOME

No vehicle is available for the selected branch and period.

→ The system returns no available vehicles.

Possible quality link: QPR-01
→ 95% of searches should finish within 2 seconds under the defined test conditions.

In Task 7 you may add or drop links depending on your analysis

User-oriented scenario - one important interaction

A scenario puts selected requirements into user context. It complements the baseline; it does not replace requirement IDs.

Example 3 — Mark a vehicle as under maintenance

USER + GOAL

Car Rental:
A customer wants to find vehicles available at a selected branch for a selected rental period.

Link: FR-03
(The system shall allow authorised branch staff to mark a vehicle as under maintenance.)

NORMAL OUTCOME

The user is authorised.

→ The vehicle is marked as under maintenance.

NON-NORMAL OUTCOME

The user is not authorised.

→ The maintenance status is not changed.

Task 6 requires ONE scenario with a normal outcome + ONE failure OR boundary outcome.

Functional behavior vs quality

Functional requirement - What should happen?
FR-01
The system shall search available vehicles for a selected branch and rental period.

Quality requirement - How well should that behavior be delivered?
QPR-01
95% of searches finish within 2 seconds with 100 concurrent users and 1,000 vehicle records.

So:
Scenario: customer searches for a vehicle.
FR-01: defines the expected **function**.
QPR-01: defines the expected **performance** of that function.

User-oriented scenario usually centers on an interaction/function, while NFRs provide measurable quality conditions around that interaction.

Traceability & objective coverage - why does each requirement exist?

M01.5 OBJECTIVE

What outcome are we trying to improve?

REQUIREMENT

What obligation supports that outcome?

SCENARIO / ACCEPTANCE

How should the behaviour be observed?

PLANNED EVIDENCE

Which future model / test may provide evidence?

Mark future evidence as PLANNED. Do not pretend a model or test already exists.

COVERAGE CHECK

Every requirement should support ⩾1 relevant M01.5 objective, and BOTH M01.5 objectives should have relevant requirement coverage.

Requirement quality review - validate before you build on it

CLEAR
Could two readers interpret it differently?

FOCUSED
Does it express one main obligation?

VERIFIABLE
Can you describe observable pass / fail evidence?

CONSISTENT
Does it contradict another requirement?

JUSTIFIED
Is the objective/source link defensible?

COMPLETE ENOUGH
Is an important condition, outcome or exception missing?

Car Rental - review and revision example

BEFORE

“The system should quickly prevent booking conflicts and notify staff.”

REVIEW FINDINGS

- “quickly” is not measurable
- two obligations are combined
- expected outcome is unclear
- objective link must be justified

AFTER

FR-03: The system shall reject a confirmed reservation request when it overlaps an existing confirmed reservation for the same vehicle.

NFR-02: Conflict checking shall complete within the assumed target under the defined test condition.

Keep stable IDs when revising wording. Record what changed and why.

Checkpoint 1 - upload your project ZIP to Moodle

WHAT TO UPLOAD

Export the COMPLETE QuickBite project ZIP after Session 04.

Upload one team ZIP to the Moodle Checkpoint 1 activity.

Check exported .zip’s name:
QuickBite_<TeamID>_after_S04_<date>.zip

Keep another copy outside the university computer.

Announcements of the lecturer
How to get a credit
Class attendance
Materials
QuickBite Project - Team IDs
QuickBite Project
Activities | Tasks
Exams | Credits
Checkpoint 1 - M01 + M02
Checkpoint 1 - M01 + M02
Presentation - M01-S01
Presentation - M01-S02
Presentation - M02-S03
Topic 2
Topic 3
Technical support
Instrukcje Moodle
Ankieta archiwizacyjna
Ankieta archiwizacyjna

Checkpoint 1 — M01 + M02 - complete by February 17th, 2027

Please upload your **latest complete QuickBite project ZIP**.

Before submitting, make sure these files contain all required answers:

- `workspace/reports/M01.md`
- `workspace/reports/M02.md`

Do not upload only `M01.md` or `M02.md`. Submit the complete project ZIP.

M01 - Required deliverables

Make sure you have completed:

- **Task 1:** Separate facts, assumptions, constraints and open questions from the QuickBite brief.
- **Task 2:** Write the problem statement and project vision.
- **Task 3:** Identify relevant stakeholders / external organizations or services. Record their needs, concerns, questions and at least one conflict.
- **Task 4:** Define your team's in-scope capabilities and explicit out-of-scope choices. Justify one debatable boundary.
- **Task 5:** Write two measurable business objectives.
- **Task 6:** Compare two lifecycle approaches and choose one for QuickBite.
- **Task 8:** Prepare a short initial feasibility assessment and identify three risks.

M02 - Required deliverables

Make sure you have completed:

- **Task 2:** Create a requirements register for your chosen scope.
- **Task 6:** Describe one important user interaction.
- **Task 7:** Review objective coverage and requirement quality.

Before uploading

- Save all answers in the workbench.
- Export the **complete project ZIP**.
- Keep another copy outside the university computer.

End of class - before you leave

- [ ] Task 6 saved: one scenario + normal + failure/boundary outcome + requirement IDs
- [ ] Task 7 saved: both M01.5 objectives reviewed + clarity/testability checked
- [ ] Any justified requirement revisions saved
- [ ] Complete project ZIP downloaded
- [ ] Checkpoint 1 ZIP uploaded to Moodle **(deadline: February 17th, 2027)**
- [ ] Backup stored outside the university computer

Next: M03 - Use cases and activity modelling

THANK YOU